Project Requirements

Project Title: Adaptive Multi-Agent Path Planning for Autonomous Vehicles

The following are the 14 major functional requirements that will be developed and implemented in our project.

1. Vehicle-to-Vehicle (V2V) Communication

- Establish communication between nearby autonomous vehicles.
- Enable vehicles to exchange traffic information, accident alerts, obstacles, and route updates.

2. Multi-Agent Autonomous Vehicle Environment

- Develop multiple independent autonomous vehicle agents.
- Each vehicle will have its own destination, speed, route, and decision-making capabilities.

3. Real-World Map Integration

- Integrate OpenStreetMap to obtain real-world road networks.
- Support actual roads, intersections, lanes, and navigation routes.

4. Traffic and Pedestrian Simulation

- Simulate realistic traffic conditions and vehicle movements.
- Include traffic signals, congestion, pedestrian crossings, and unexpected road conditions.

5. Motion Planning and Performance Benchmarking

- Implement and compare different path-planning algorithms.
- Evaluate travel time, collision rate, route efficiency, safety, and communication performance.

6. Decentralized Vehicle Communication

- Enable vehicles to communicate directly without depending on a central communication controller.
- Support independent information sharing and coordination between vehicles.

7. Adaptive Path Planning

- Implement dynamic route planning using algorithms such as A*.
- Automatically calculate alternative routes during traffic congestion, accidents, or road blockages.

8. AI-Based Decision-Making

- Develop an intelligent decision-making system for autonomous vehicles.
- Enable vehicles to independently accept or reject route suggestions based on traffic conditions, safety, and travel efficiency.

9. Collision Detection and Avoidance

- Detect nearby vehicles, pedestrians, and obstacles.
- Implement collision avoidance mechanisms and maintain safe distances.

10. Emergency Vehicle Priority

- Identify emergency vehicles such as ambulances.
- Enable nearby vehicles to safely adjust their movement and provide priority.

11. Communication Range Detection

- Implement configurable vehicle communication ranges.
- Automatically establish communication when vehicles enter the specified communication radius.

12. Communication Failure Handling

- Detect communication failures, delayed messages, and packet loss.
- Implement fallback mechanisms to maintain safe vehicle operation during communication failures.

13. AI Model Training Using Reinforcement Learning

- Develop and train a reinforcement learning model for intelligent route selection.
- Improve vehicle decision-making through interaction with simulated traffic environments.

14. Real-Time 3D Visualization and Interactive Dashboard

- Develop a web-based dashboard using Django and Three.js.
- Visualize realistic roads, buildings, autonomous vehicles, pedestrians, and traffic conditions.
- Display real-time vehicle movements, V2V communication, route changes, and AI decisions.

Project Objective: To develop an intelligent, decentralized multi-agent autonomous vehicle simulation system that enables real-time V2V communication, adaptive route planning, AI-based decision-making, and safe navigation in realistic traffic environments.